Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3301-DE01

Title: (BEng/MEng)

Structures, Civil Materials And Geomatics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* • *In line with the Department of Engineering document entitled "Student Guidance on the Use of Generative Artificial Intelligence and Related Technology for the Department of Engineering" policy in assessments, the use of generative artificial intelligence tools and related technologies is strictly prohibited for this exam.*

Revision:	

Question 1

(a) A fixed-end beam of length L with a uniform vertical load w per unit length covering its entire length.

(i) Calculate the degree of indeterminacy.

[10%]

(ii) Estimate the maximum sagging moment using the approximate method for analysing vertically loaded indeterminate frames.

[20%]

(b) The cable in Figure Q1 will break when the maximum tension reaches 100 kN. Determine the maximum uniform distributed load w required to develop this maximum tension.

[70%]

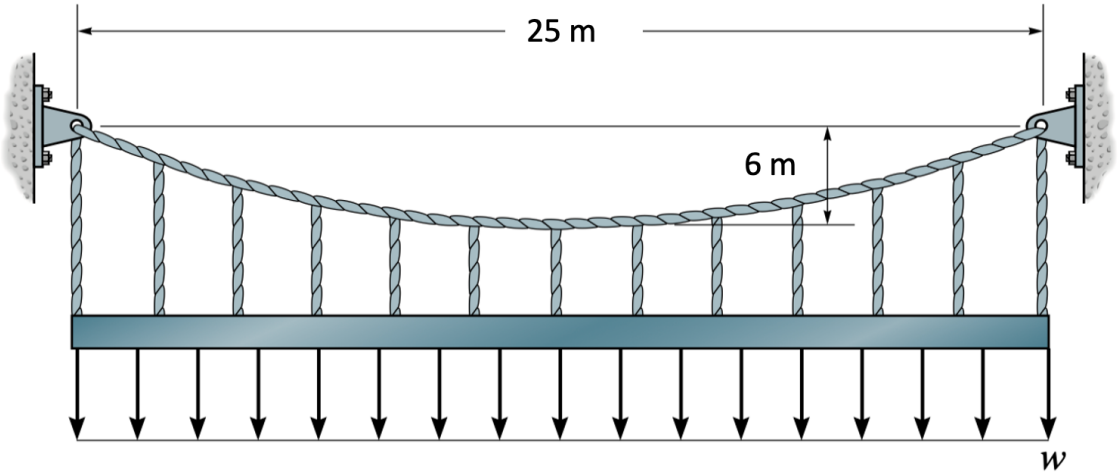


Figure Q1

Question 2

(a) Consider the continuous beam in Figure Q2a with three simple supports. Assume that the material and section properties are constant throughout the length of the beam and neglect axial loads and deformation.

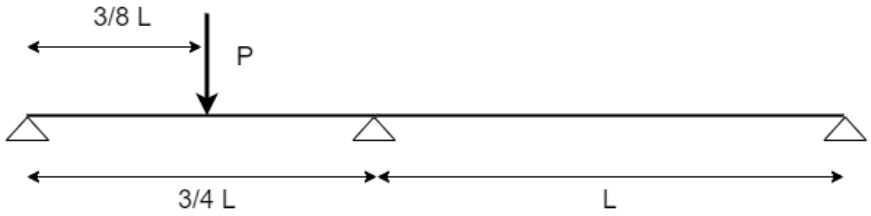


Figure Q2a

(i) Sketch the expected mode of failure, and calculate the value of the point load P_u that would lead to the collapse of the beam due to the formation of plastic hinge(s) as a function of the plastic moment (M_p). [35%]

(ii) Based on the assumed mode of failure for *Part (a)* estimate the support reactions and sketch the quantitative (i.e., labeling all minima and maxima) shear force and bending moment diagrams for the beam at the point of collapse as a function of the plastic moment (M_p). [30%]

(b) Consider the frame shown in Figure Q2b with fixed supports at the bottom, rigid connections between each beam and column, and vertical and horizontal concentrated loading as shown. Assume that the material and section properties are constant throughout. Describe, including sketches, four possible elementary collapse mechanisms and explain the procedure to determine the critical loading based on these collapse mechanisms. [35%]

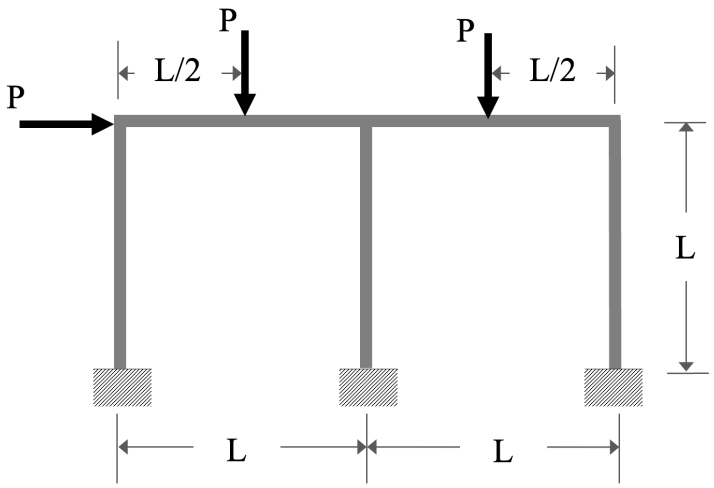


Figure Q2b

continued

Question 3

(a) Determine the mix quantities (water, cement, fine aggregate and coarse aggregate) for a $0.2m^3$ trial concrete mix with a characteristic strength of 40MPa at 28 days (based upon 2% defectives, 20 specimens tested) and a target slump of 30mm. The following materials should be used.

- Cement: OPC (EN Class 42.5)
- Coarse aggregate: 20mm crushed
- Fine aggregate: uncrushed, 40% passing 0.6mm

[60%]

(b) In masonry bondwork, what is the difference between a running bond and an English bond? What are the advantages and disadvantages of these types of bondwork relative to each other?

[20%]

(c) In timber construction, what are the relative advantages and disadvantages of Oriented Strand Board (OSB) over Particle Board (Chipboard).

[20%]

continued

Question 4

(a) In a large-scale road project, an engineering apprentice is tasked with vertical control to monitor pavement settlements during construction. While the apprentice recognises that both an auto level and a total station can accomplish vertical control, they are uncertain about which is the most viable option for the job. As project lead, foreseeing the requirements and challenges of the work, provide a detailed explanation to guide the choice of the appropriate equipment. [30%]

(b) A link survey was conducted from Station A to Station B, passing through four traverse stations (1, 2, 3, and 4). The horizontal clockwise angles and distances between these stations are provided. The co-ordinates for Stations A and B, along with their respective control stations R1 and R2, are also provided. Based on this data, construct a traverse diagram. Using the given information, calculate the coordinates of Stations 1 to 4, determine the misclosures, calculate error and adjust the co-ordinates using the equal distribution method. Finally, comment on the overall quality of the survey conducted. [70%]

Horizontal Clockwise Angles	
AT-FROM-TO	Angle
A-R1-1	115^0 37’ 17”
1-A-2	168^0 19’ 11”
2-1-3	281^0 12’ 40”
3-2-4	242^0 53’ 39”
4-3-B	80^0 26’ 24”
B-4-R2	173^0 31’ 03”

Horizontal Distance	
FROM-TO	Distance (m)
A-1	83.346
1-2	77.345
2-3	130.634
3-4	123.715
4-B	89.896

continued

Known Co-ordinates		
Control Station	Easting (m)	Northing (m)
Reference point - R1	1764.507	2104.938
Reference point – R2	2198.076	1884.945
Station A	1875.369	2043.008
Station B	2101.624	1904.041

END